

Computer Networking: Security (CSE 3752)

CNS Assignment – 1

Q.1. List out the common categories associated with the security management of an organization. With the help of suitable examples, briefly explain how technical control improves the security performance.

Q.2. In your computing system, you are advised to update the operating system for security concerns. Do you go for this? During updating, if the system crashes, then to overcome this, you are suggested to go back to the previous state of the operating system. Compare and analyse the control types applicable in the above scenarios.

Q.3. A healthcare organization stores patient records online. Recently, unauthorized access was detected, and some records were modified. Explain which components of the CIA triad are violated and suggest suitable measures.

Q.4. State the importance of non-repudiation and discuss its key aspects briefly in a digital communication environment.

Q.5. An organization uses biometric login systems; explain how authentication methods enhance security compared to passwords.

Q.6. Justify the principle of “never trust, always verify” from the Zero Trust model.

Q.7. Explain the use of the following techniques associated with deception and disruption to strengthen security in computer networking.

(i) Honeypot (ii)Honeynet (iii)Honeyfile (iv)Honeytoken

Q.8. In a startup environment, developers are allowed to directly implement changes in the production system without defined responsibility. Evaluate the associated risks and explain how ownership and accountability improve system security.

Q.9. A person received two encrypted messages from their financial advisor. In one message, the same letters from the original message appear, but in different positions, while in another message, completely different letters appear. Analyze both cases and determine which cryptographic techniques are used in the above cases. Categorize them as symmetric/asymmetric.

Q.10. Decrypt the ciphertext “**ROTI**” using the Caesar cipher with key 6. Why is the Caesar Cipher considered insecure in modern cryptography?

Q.11. Encrypt the plaintext "TOMORROW" using the Playfair cipher with the key "SECRET". Assume the letter I & J shares the same cell in the key matrix.

Q.12. What will be the plain text corresponding to cipher text “DYRYVGKP” if Vigenère cipher is used with the keyword as “KEY”?

Q.13. A bank wants to securely transmit a 64-bit transaction record $(0110102341101023)_{\text{Hex}}$ between two branches. Before sending, the bank encrypts the data using DES with a secret key.

- Convert the given hexadecimal transaction record into binary.
- Apply the Initial Permutation (IP) table of DES and show the resulting binary sequence.
- Explain why the initial permutation is important in the DES encryption process.

Q.14. In the DES function (f), after the S-Boxes, you got a 32-bit output. Suppose that the output is $0x00000001$ (only the very last bit is 1). Apply the **P-box permutation** to this 32-bit block. Determine the output in both Binary and Hexadecimal.

P-Table Reference:

$P = [16, 7, 20, 21, 29, 12, 28, 17, 1, 15, 23, 26, 5, 18, 31, 10, 2, 8, 24, 14, 32, 27, 3, 9, 19, 13, 30, 6, 22, 11, 4, 25]$

Q.15. State the significance of the S box in DES? Given the elements of the 2nd row (i.e. row no. "01") in an S box of DES as 14,4,13,1,2,15,11,8,3,10,6,12,5,9,0 & 7. If the 6-bit input to the S box is "011011", determine the corresponding 4-bit output.

Q.16. Given the 56-bit input to generate the key for round 3 in DES as $0X123456789ABCDE$. Find the 56-bit input to generate the key for round 4 in DES.

Q.17. Consider the following state matrix after SubByte transformation in Round 10 of AES 128 algorithm.

D4	E0	B8	1E
BF	B4	41	27
5D	52	11	98
30	AE	F1	E5

- Apply the ShiftRows operation and determine the resulting state matrix.
- Find the output state matrix after Add Round Key transformation if the 128 bit round key is given as $(000102030405060708090A0B0C0D0E0F)_{\text{Hex}}$.

Q.18. Using RSA Algorithm, compute ciphertext C for $n=33$, $e=3$ and plaintext $M=4$. Also find out the respective decryption key "d".

Q.19. Alice and Bob agree to use the prime number $p=17$ and the generator $g=5$ for their key exchange using Diffie-Hellman algorithm. Alice sends Bob $a=2$. Bob chooses the random number $b=3$. What is Alice and Bob's shared secret key?

Q.20. Explain the process of creating and verifying a digital signature. Where are digital signatures used in real life?

Q.21. State the significance of Certificate Revocation Lists (CRLs) and Online Certificate Status Protocol (OCSP) associated with the certificate validity process, ensuring a secure interaction over the internet.

Q.22. A tech company's confidential prototype details appear in a rival's new product. There is no sign of external breach, but one employee recently resigned. Name and explain the type of threat the scenario represents.

Q.23. What is meant by vulnerable software, and explain how the vulnerabilities occur.

Q.24. A trusted software update from a vendor introduces malware into multiple organisations simultaneously. Evaluate why attackers choose supply chain attacks instead of direct attacks. What advantages do they gain logically and strategically?

Q.25. Briefly explain the following techniques associated with attacks made using human psychology.

(i) Phishing (ii) Misinformation (iii) Brand impersonation (iv) Typosquatting